

MA1513 Exam
AY2022/2023 Semester I
(Answer Key)

Question #: 1

$$\left(\begin{array}{ccccc|c} 1 & 2 & 1 & 0 & 2 & a \\ 0 & 0 & 1 & 0 & 1 & b \\ 0 & 0 & 2 & 0 & 2 & c \end{array} \right)$$

- A. By further reducing the third row of the augmented matrix, we get $(0 \ 0 \ 0 \ 0 \ 0 \mid c - 2b)$. This implies the system is only consistent when $c = 2b$.
B. The REF of the matrix has three non-pivot columns (2, 4 and 5). So there are 3 parameters.
C. In view of B above, it is not possible for the system to have exactly one solution.
D. If $c - 2b$ is not zero, the last column will be a pivot column.
E. If $c = 0$, but b is not zero, the system will have no solution.

The above is an augmented matrix of a linear system with a, b, c being the constant terms. Which of the following statements are true? (Pick all correct options)

- A. This system is consistent only when $c = 2b$.
B. If the system is consistent, the general solution has two parameters.
C. For some values of a, b, c , the system has exactly one solution.
D. The last column (constant terms) of a row echelon form of the augmented matrix can be a pivot column.
E. If $c = 0$, the system will have infinitely many solutions.

Item Weight: 3.0

Question #: 2

Given that $\mathbf{A}$ is a singular $n \times n$ matrix, which of the following statements are true? (Pick all correct options)

- A. The rank of $\mathbf{A}$ is n .
B. The determinant of $\mathbf{A}$ is 0.
C. The homogeneous system $\mathbf{Ax} = \mathbf{0}$ does not have a trivial solution.
D. The n column vectors of $\mathbf{A}$ form a linearly dependent set.
E. There is a non-zero vector in the nullspace of $\mathbf{A}$.
- A. If $\mathbf{A}$ is singular, $\text{rank}(\mathbf{A}) < n$.
B. All singular matrices have determinant 0.
C. Every homogeneous system has the trivial solution, regardless of $\mathbf{A}$ is singular or not.
D. Since $\det(\mathbf{A}) = 0$ (from B), the n columns vectors are linearly dependent.
E. Since the homogeneous system $\mathbf{Ax} = \mathbf{0}$ has non-trivial solution, this means the nullspace of $\mathbf{A}$ contains non-zero vectors.

Item Weight: 3.0

Question #: 3

$$\begin{vmatrix} a & b & c \\ a & 0 & c \\ * & * & * \end{vmatrix} = 0$$

The left-hand side above is a 3 x 3 determinant, where a, b, c are non-zero distinct numbers. Which of the following can be the third row of the determinant, regardless of the values of a, b, c? (Pick all correct options)

- A. (a b c)
- B. (0 p 0) where p is a non-zero number.
- C. (a n c) where n is not equal to 0 and b.
- D. (a a a)
- E. (c b a)

- A. The first and third rows are identical, so $\det = 0$
- B. The first column and third column are scalar multiple of each other (linearly dependent), so $\det = 0$
- C. The first column and third column are scalar multiple of each other (linearly dependent), so $\det = 0$
- D. there are values for a, b, c such that the resulting matrix has non-zero determinant
- E. Same as D.

Item Weight: 3.0

Question #: 4

Which of the following linear spans do not contain the vector (1,2,3,4) ? (Pick all correct answers)

- A. $\text{Span}\{ (4,3,2,1) \}$
- B. $\text{Span}\{ (1,2), (3,4) \}$
- C. $\text{Span}\{ (1,0,3,0), (0,2,0,4) \}$
- D. $\text{Span}\{ (2,3,4,1), (3,4,1,2), (4,1,2,3) \}$
- E. $\text{Span}\{ (4,0,0,0), (0,3,0,0), (0,0,2,0), (0,0,0,1) \}$

- A. This span contains all scalar multiples of (4,3,2,1).
- B. This span contains vectors from $\mathbb{R}^2$
- C. $(1,2,3,4) = (1,0,3,0) + (0,2,0,4)$ is a linear combination of the two vectors in the span
- D. Check that (1,2,3,4) is not a linear combination of the three vectors in the span (e.g. use Gaussian elimination)
- E. $(1,2,3,4) = 1/4(4,0,0,0) + 2/3(0,3,0,0) + 3/2(0,0,2,0) + 4(0,0,0,1)$ is a linear combination of the four vectors in the span

Item Weight: 3.0

Question #: 5

Which of the following sets of vectors in $\mathbb{R}^3$ all lie on the same plane? (Pick all correct answers)

- A. (0, 0, 0), (1, 2, 0), (2, 0, 1)
- B. (1, 2, 1), (2, 1, 1), (1, 1, 2)
- C. (1, 2, 3), (4, 5, 6), (7, 8, 9)
- D. (2, 2, 0), (0, 0, 3), (1, 1, 1), (3, 3, 2)
- E. (1, 0, 1), (0, 1, 1), (0, 0, 1)

- Check whether the set contains at most 2 linearly independent vectors:
- A. Since (0,0,0) is obviously "redundant", there are only 2 linearly independent vectors.
 - B. Check directly the three vectors are linearly independent.
 - C. Check directly the three vectors are linearly dependent.
 - D. Since there are 4 vectors, it is not enough to check that the set is linearly dependent. We check $(1,1,1) = 1/2(2,2,0) + 1/3(0,0,3)$. So (1,1,1) is redundant; $(3,3,2) = 3/2(2,2,0) + 2/3(0,0,3)$. So (3,3,2) is also redundant. So there are only two linearly independent vectors.
 - E. Check directly the three vectors are linearly independent.

Item Weight: 3.0

Question #: 6

The projection of the vector $(1,2,3,0)$ onto the subspace $V = \text{span}\{ (1,0,1,0), (0,1,0,1) \}$ is given by $\mathbf{p} = (a, b, c, d)$ where $a = \underline{1}$, $b = \underline{2}$, $c = \underline{3}$, $d = \underline{4}$

1. 2
2. 1
3. 2
4. 1

Let A be the 4×2 matrix $\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$ and b the 4×1 column $\begin{bmatrix} 1 \\ 2 \\ 3 \\ 0 \end{bmatrix}$
Find the least squares solution x_0 of $Ax = b$ (by solving $A^T A x = A^T b$):
 $x_0 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$
Then $p = Ax_0 = \begin{bmatrix} 2 \\ 1 \\ 2 \\ 1 \end{bmatrix}$

Item Weight: 3.0

Question #: 7

Let $\mathbf{v}$ be a vector in $\mathbf{R}^3$ and $S = \{ (1,1,0), (1,0,1), (0,1,1) \}$ and $T = \{ (3,0,0), (0,4,0), (0,0,5) \}$ be two bases for $\mathbf{R}^3$. Suppose the coordinate vector of $\mathbf{v}$ with respect to S is $[\mathbf{v}]_S = (1,2,3)$. The coordinate vector of $\mathbf{v}$ with respect to T is $[\mathbf{v}]_T = (a, b, c)$ where $a = \underline{1}$, $b = \underline{2}$, $c = \underline{3}$

1. 1
2. 1
3. 1

$\mathbf{v} = 1(1,1,0) + 2(1,0,1) + 3(0,1,1) = (3, 4, 5)$
Solve $(3,4,5) = a(3,0,0) + b(0,4,0) + c(0,0,5)$, we get
 $a = 1, b = 1$ and $c = 1$
So $[\mathbf{v}]_T = (1, 1, 1)$

Item Weight: 3.0

Question #: 8

Let A be a 3×5 matrix. Which of the following statements are true? (Pick all correct options)

- A. The row space of A is a subspace of $\mathbf{R}^3$.
- B. The sum of two columns of A is again a column of A .
- C. A cannot be a full rank matrix.
- D. The nullity of A is at least 2.
- E. The nullspace of A is a subspace of $\mathbf{R}^5$.

A. Since A has 5 columns, the row space is a subspace of $\mathbf{R}^5$
B. The sum of two columns of A is in the column space of A , but need not be a column of A itself.
C. The rank of A can be 3, and hence can be full rank.
D. Since there are 5 columns, and the rank of A is at most 3, it means the nullity of A is at least 2
E. Since A has 5 columns (correspond to 5 variables in the homogeneous system $Ax = 0$), the solutions are vectors in $\mathbf{R}^5$. Hence the nullspace of A is a subspace of $\mathbf{R}^5$.

Item Weight: 3.0

Question #: 9

Suppose $T: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ is a linear transformation and we are given

$$T\left(\begin{pmatrix} 2 \\ 3 \end{pmatrix}\right) = \begin{pmatrix} 1 \\ -2 \end{pmatrix} \text{ and } T\left(\begin{pmatrix} 3 \\ 4 \end{pmatrix}\right) = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

$$A[2;3] = [1;-2] \text{ and } A[3;4] = [2;1]$$

By stacking

$$A[2 \ 3; 3 \ 4] = [1 \ 2; -2 \ 1].$$

$$\text{Then } A = [1 \ 2; -2 \ 1] [2 \ 3; 3 \ 4]^{-1} = [2 \ -1; 11 \ 8]$$

The standard matrix of T is given by $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

where $a = \underline{\quad 1 \quad}$, $b = \underline{\quad 2 \quad}$, $c = \underline{\quad 3 \quad}$, $d = \underline{\quad 4 \quad}$.

1. 2
2. -1
3. 11
4. 8

Item Weight: 3.0

Question #: 10

$$A = \begin{pmatrix} 2 & * & * \\ 0 & 3 & * \\ 0 & 0 & 4 \end{pmatrix}, B = \begin{pmatrix} 2 & * & * \\ 0 & 2 & * \\ 0 & 0 & 3 \end{pmatrix}, C = \begin{pmatrix} 2 & * & * \\ 0 & 2 & * \\ 0 & 0 & 2 \end{pmatrix}$$

Let A, B, C be the above three upper triangular matrices where each $*$ can represent any number. Which of the following statements are true? (Pick all correct answers)

- A. Matrix A may or may not be diagonalizable.
- B. Matrix B may or may not be diagonalizable.
- C. Matrix C must be non-diagonalizable.
- D. All the eigenspaces of matrix A must have dimension 1.
- E. Matrix B must have an eigenspace with dimension 2.

- A. A has three distinct eigenvalues, it must be diagonalizable
- B. B has only two eigenvalues, which may or may not be diagonalizable, depending on the non-diagonal entries.
- C. If C is a scalar matrix, it will be diagonalizable
- D. Since A has three distinct eigenvalues, their multiplicity are all equal to 1. Hence the dimension of the eigenspaces are all equal to 1.
- E. Both eigenspaces E_2 and E_3 of B can be of dimension 1.

Item Weight: 3.0

Question #: 11

Let P be a non-singular 3×3 matrix with columns C_1 , C_2 and C_3 in that order. Suppose

$$P^{-1}AP = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Which of the following statements are true?

A. The nullspace of A is $\text{span}\{C_2\}$.

B. $C_1 + C_2$ is an eigenvector of A .

C. $A^2 = A$

D. The matrix with columns $3C_1$, $-C_2$, $2C_3 + C_1$ will diagonalize matrix A .

E. A is a non-singular matrix.

A. The nullspace of A is the same as the eigenspace corresponding to eigenvalue 0. So it is span by the second column C_2 of P .

B. C_1 and C_2 correspond to different eigenvalues. The sum is not an eigenvector.

C. $A = PDP^{-1}$ and $A^2 = P(D^2)P^{-1}$. In this case, $D^2 = D$. So $A^2 = A$.

D. $3C_1$ and $2C_3 + C_1$ are still eigenvectors corresponding to eigenvalue 1, and $-C_2$ is still an eigenvector corresponding to eigenvalue 0. The three new columns are linearly independent. So the new matrix will diagonalize A .

E. As A has eigenvalue 0, it is singular.

Item Weight: 3.0

Question #: 12

A certain 3rd order ODE is given by $y''' + ay'' + by' + cy = 0$. This ODE can be converted to a 3×3 SDE given by

$$y' = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & -1 & 3 \end{pmatrix} y.$$

The three entries 2, -1, 3 in the last row of the underlying matrix correspond to $-c$, $-b$, $-a$ in the coefficient of the ODE.

Then $a = \underline{\quad 1 \quad}$, $b = \underline{\quad 2 \quad}$, $c = \underline{\quad 3 \quad}$.

1. -3

2. 1

3. -2

Item Weight: 3.0

Question #: 13

An SDE $\mathbf{y}' = \mathbf{A}\mathbf{y}$ gives an equilibrium point which is a degenerate node. Suppose the matrix $\mathbf{A}$ has an eigenvalue 3 with an eigenvector $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ and corresponding generalised eigenvectors $\begin{pmatrix} 1/3 \\ s \end{pmatrix}$ for any s . Which of the following are solutions of the SDE? (Pick all correct answers)

A.

$$\begin{pmatrix} 0 \\ 3 \end{pmatrix} e^{3t}$$

The general solution of the SDE is given by $c_1 [0; 1]e^{3t} + c_2 ([0; t] + [1/3; s]) e^{3t}$ where c_1 , c_2 and s can be any numbers.

B.

$$\begin{pmatrix} 1 \\ t \end{pmatrix} e^{3t}$$

- A. This is a solution of the SDE by taking $c_1 = 3$ and $c_2 = 0$
- B. We cannot get this vector function with any value of c_1 , c_2 and s (the first component in the vector must be $1/3$)
- C. We cannot get this vector function with any value of c_1 , c_2 and s (the second component in the vector must contain t)
- D. This is a solution of the SDE by taking $c_1 = 0$, $c_2 = 1$ and $s = 1$
- E. Same as C.

C.

$$\begin{pmatrix} 1/3 \\ 0 \end{pmatrix} e^{3t}$$

D.

$$\begin{pmatrix} 1/3 \\ t + 1 \end{pmatrix} e^{3t}$$

E.

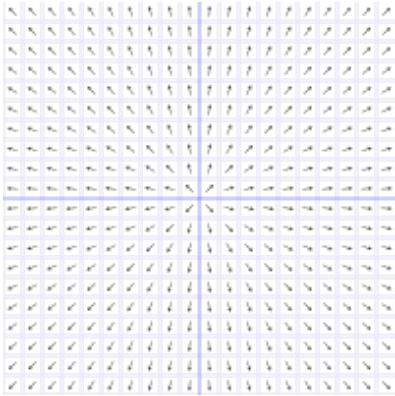
$$\begin{pmatrix} 1/3 \\ 1 \end{pmatrix} e^{3t}$$

Item Weight: 3.0

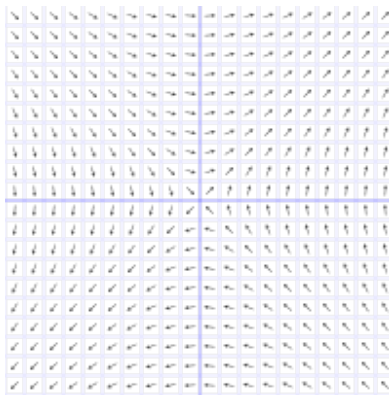
Question #: 14

Which of the following direction fields belong to the SDE $y' = \begin{pmatrix} 2 & 4 \\ 1 & 2 \end{pmatrix} y$?

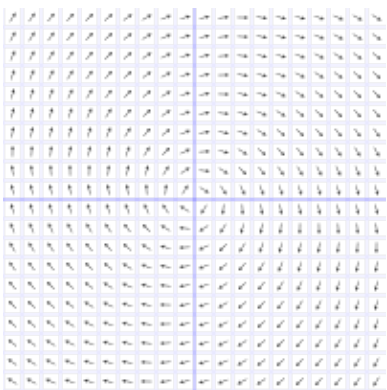
A.



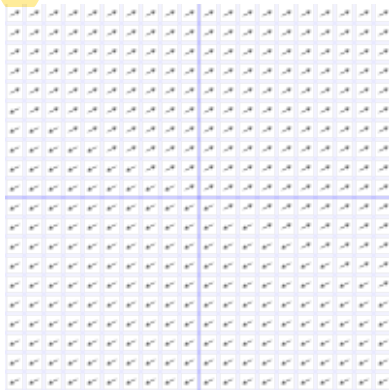
B.



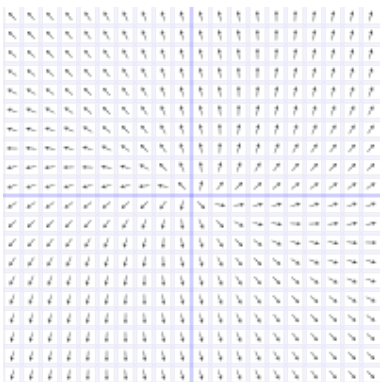
C.



D.



E.



The underlying matrix $\begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}$ has eigenvalue 0 and 4.
So the phase portrait consists of trajectories which are parallel straight lines. The direction field of D gives the right portrait.
A is a star node
B is a saddle point
C is a center
E is a spiral point

Item Weight: 3.0

Question #: 15

Suppose a 2 x 2 SDE $\mathbf{y}' = \mathbf{A}\mathbf{y}$ has the following fundamental set of solutions:

$$X_1 = e^t \begin{bmatrix} 0 \\ 1 \end{bmatrix} \cos 3t + \begin{bmatrix} 1 \\ 0 \end{bmatrix} \sin 3t, X_2 = e^t \left[\begin{bmatrix} 0 \\ 1 \end{bmatrix} \sin 3t - \begin{bmatrix} 1 \\ 0 \end{bmatrix} \cos 3t \right]$$

Which of the following statements are true? (Pick all correct answers)

- A. The equilibrium point of the SDE is unstable.
- B. The matrix $\mathbf{A}$ has eigenvalues $1+3i$ and $1-3i$.
- C. The matrix $\mathbf{A}$ has an eigenvector $\begin{pmatrix} 1 \\ i \end{pmatrix}$.
- D. The equilibrium point is a center.
- E. The trajectories of the SDE are counter-clockwise.

The given solutions suggest that $\mathbf{A}$ has a complex eigenvalue $1 + 3i$ with corresponding eigenvector $\begin{bmatrix} 0; 1 \end{bmatrix} + \begin{bmatrix} -1; 0 \end{bmatrix} i = \begin{bmatrix} -i; 1 \end{bmatrix}$

A. Since the real part of the eigenvalue is positive, the e.p. is unstable

B. The complex eigenvalues come in conjugate pairs $1+3i$ and $1-3i$

C. $\begin{bmatrix} 1; i \end{bmatrix} = i \begin{bmatrix} -i; 1 \end{bmatrix}$ which is a scalar multiple of an eigenvector.

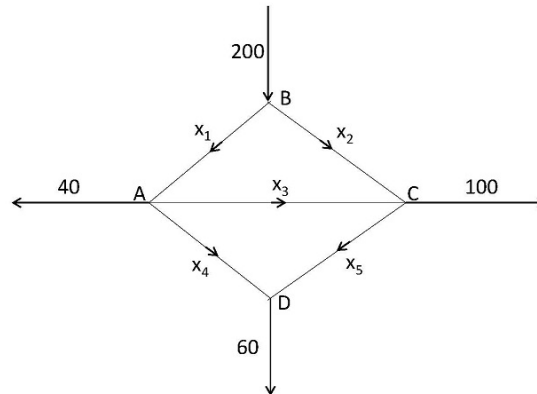
D. The real part of eigenvalue is not 0, so the e.p. is a spiral instead of a center

E. The matrix $\mathbf{A} = \mathbf{PDP}^{-1} = \begin{bmatrix} 1 & 3; -3 & 1 \end{bmatrix}$. Testing the point $(1,0)$ gives the tangent vector $\begin{bmatrix} 1; -3 \end{bmatrix}$ which is pointing downward. So the trajectories are spiralling clockwise.

Item Weight: 3.0

LQ1

The network in the diagram shows the traffic flow (in vehicles per hour) over several one-way streets in the city area during peak hours on a typical weekday.



- [3 marks] Set up a linear system $\mathbf{Ax} = \mathbf{b}$ with 4 equations (in the order of the junctions A, B, C and D) and traffic volumes arranged in the order of x_1, x_2, x_3, x_4 and x_5 . Write down the matrix $\mathbf{A}$ and vector $\mathbf{b}$ explicitly. All entries of $\mathbf{b}$ should be positive.
- [3 marks] Write down the general solution of the linear system in (a). (You may use MATLAB for the working.)
- [2 marks] What is the largest possible value for x_3 , given that all the x_i must be non-negative?
- [2 marks] If the street between A and D is closed, what is the range for the possible values of x_1 ?

(a) Equating the inflow and outflow at each junction, we have:

$$\begin{aligned} A : \quad x_1 &= x_3 + x_4 + 40 \\ B : \quad 200 &= x_1 + x_2 \\ C : \quad x_2 + x_3 &= x_5 + 100 \\ D : \quad x_4 + x_5 &= 60 \end{aligned}$$

which gives the linear system:

$$\begin{aligned} x_1 &\quad - x_3 - x_4 &= 40 \\ x_1 + x_2 &&= 200 \\ x_2 + x_3 &\quad - x_5 &= 100 \\ &x_4 + x_5 &= 60 \end{aligned}$$

The matrix form of the linear system is $\mathbf{A}\mathbf{x} = \mathbf{b}$ where

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & -1 & -1 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 40 \\ 200 \\ 100 \\ 60 \end{pmatrix}$$

(b)

$$\left(\begin{array}{ccccc|c} 1 & 0 & -1 & -1 & 0 & 40 \\ 1 & 1 & 0 & 0 & 0 & 200 \\ 0 & 1 & 1 & 0 & -1 & 100 \\ 0 & 0 & 0 & 1 & 1 & 60 \end{array} \right) \xrightarrow{G.J.E.} \left(\begin{array}{ccccc|c} 1 & 0 & -1 & 0 & 1 & 100 \\ 0 & 1 & 1 & 0 & -1 & 100 \\ 0 & 0 & 0 & 1 & 1 & 60 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

Let $x_3 = s, x_5 = t$. Then $x_4 = 60 - t, x_2 = 100 + t - s, x_1 = 100 - t + s$.

The general solution of the system is:

$$\begin{cases} x_1 = 100 - t + s \\ x_2 = 100 + t - s \\ x_3 = s \\ x_4 = 60 - t \\ x_5 = t \end{cases}$$

(c) From $x_4 = 60 - t \geq 0$, we have $t \leq 60$.

From $x_2 = 100 + t - s \geq 0$, we have $s \leq 100 + t \leq 160$.

So the largest possible value for x_3 is 160.

(d) Since the street between A and D is closed, we have $x_4 = 0$ which implies $t = 60$.

Then $x_1 = 100 - 60 + s = 40 + s$.

Since $0 \leq s \leq 160$, this implies $40 \leq x_1 \leq 200$.

LQ2

The revenue y (in \$100. i.e. when $y = 1$, it means the revenue is \$100) from the sale of x (in 100 of units, i.e, when $x = 1$, it means 100 units are sold) of a product by company Z can be described using the equation:

$$y = mx + nx^2 \quad (*)$$

Some data points (x, y) were collected : $(1, 1.8), (2, 2.7), (3, 3.4), (4, 3.8), (5, 3.9)$.

- (a) [3 marks] Set up a linear system $\mathbf{Ax} = \mathbf{b}$ with five equations to find the coefficients m and n in the equation $(*)$. Write down the matrix $\mathbf{A}$ and vector $\mathbf{b}$ explicitly.
- (b) [3 marks] Write down a matrix equation that allows you to find the least squares solution(s) of the system $\mathbf{Ax} = \mathbf{b}$. Use it to find the values of m and n that produce the least squares fit of the given data points. (You may use MATLAB for the working. Give your answers up to 4 decimal places.)
- (c) [2 marks] What is the projected revenue when 500 units of the product are sold based on the least squares fit model in (b)? (Give your answer to the nearest dollar.)
- (d) [3 marks] According to the least squares fit model in (b), should company Z sell more than 500 units of the product to increase the revenue? Why?

(a) By substituting the 5 data points into equation (*), we have a linear system:

$$\begin{aligned} 1.8 &= m + n \\ 2.7 &= 2m + 4n \\ 3.4 &= 3m + 9n \\ 3.8 &= 4m + 16n \\ 3.9 &= 5m + 25n \end{aligned}$$

So

$$\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 2 & 4 \\ 3 & 9 \\ 4 & 16 \\ 5 & 25 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1.8 \\ 2.7 \\ 3.4 \\ 3.8 \\ 3.9 \end{pmatrix}$$

(b) The equation is given by $\mathbf{A}^T \mathbf{A} \mathbf{x} = \mathbf{A}^T \mathbf{b}$ where

$$\mathbf{A}^T \mathbf{A} = \begin{pmatrix} 55 & 225 \\ 225 & 979 \end{pmatrix}, \quad \mathbf{A}^T \mathbf{b} = \begin{pmatrix} 52.1 \\ 201.5 \end{pmatrix}.$$

$$\text{So the least squares solution is } \mathbf{x}_0 = \begin{pmatrix} 55 & 225 \\ 225 & 979 \end{pmatrix}^{-1} \begin{pmatrix} 52.1 \\ 201.5 \end{pmatrix} = \begin{pmatrix} 1.7604 \\ -0.1988 \end{pmatrix}.$$

So $m = 1.7604$, $n = -0.1988$.

(c) $y = 5m + 5^2n = 5(1.7604) + 25(-0.1988) = 3.8329$.

So the projected revenue is \$383.

(d) No. Since (*) is a quadratic equation, the value of y will decrease after the maximum value:

$$\frac{dy}{dx} = m + 2nx = 0 \Rightarrow x = -\frac{m}{2n} = -\frac{1.7604}{2(-0.1988)} = 4.4276.$$

So the maximum revenue is when $x = 4.4276$ (i.e. around 443 units). Any additional units more than 443 produced will lead to decrease in revenue.

LQ3

In a certain city, a car rental company has 3 picking up and returning locations: airport, train terminal and downtown. The weekly pattern of pick-ups and returns is given below:

- (i) 90% of the cars picked up from the airport are returned to the airport, while 9% and 1% are returned to the train terminal and downtown respectively;
- (ii) 90% of the cars picked up from the train terminal are returned to the train terminal, while 9% and 1% are returned to the airport and downtown respectively;
- (iii) 90% of the cars picked up from downtown are returned to downtown, while 9% and 1% are returned to the airport and train terminal respectively.

- (a) [3 marks] Find the 3×3 matrix $\mathbf{A}$ such that the matrix equation $\mathbf{y}_n = \mathbf{A}\mathbf{y}_{n-1}$ where

$$\mathbf{y}_n = \begin{pmatrix} a_n \\ t_n \\ d_n \end{pmatrix} \text{ represents the number of cars } a_n, t_n, d_n \text{ ready to be rented out at the three}$$

locations: airport, train terminal and downtown, at the beginning of week n .

- (b) [3 marks] Find a matrix $\mathbf{P}$ that diagonalizes the matrix $\mathbf{A}$ in part (a), and the corresponding diagonal matrix $\mathbf{D}$. (You may use MATLAB for the working. Give the entries of $\mathbf{P}$ and $\mathbf{D}$ in fractions.)
- (c) [3 marks] Use (b) to find $\mathbf{A}^n$ when n tends to infinity. Show how your answer is derived. (Give your answer in fractions.)
- (d) [3 marks] At the steady state, what is the proportion of the cars (among all the cars of the company) ready to be rented at each of the 3 locations? (Give your answers in fractions.)

(a) The required equations are given by:

$$\begin{aligned} a_n &= 0.9a_{n-1} + 0.09t_{n-1} + 0.09d_{n-1} \\ t_n &= 0.09a_{n-1} + 0.9t_{n-1} + 0.01d_{n-1} \\ d_n &= 0.01a_{n-1} + 0.01t_{n-1} + 0.9d_{n-1} \end{aligned}$$

So the matrix $\mathbf{A} = \begin{pmatrix} 0.9 & 0.09 & 0.09 \\ 0.09 & 0.9 & 0.01 \\ 0.01 & 0.01 & 0.9 \end{pmatrix}$.

(b) From MATLAB, $\mathbf{P} = \begin{pmatrix} -1 & 99/19 & 0 \\ 1 & 91/19 & -1 \\ 0 & 1 & 1 \end{pmatrix}$ and $\mathbf{D} = \begin{pmatrix} 81/100 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 89/100 \end{pmatrix}$

(c) For very large n ,

$$\mathbf{A}^n = \mathbf{P}\mathbf{D}^n\mathbf{P}^{-1} \approx \mathbf{P} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \mathbf{P}^{-1} = \begin{pmatrix} 9/19 & 9/19 & 9/19 \\ 91/209 & 91/209 & 91/209 \\ 1/11 & 1/11 & 1/11 \end{pmatrix}$$

(d) Suppose currently the number of cars in the 3 locations are a, t, d . Then total number of cars of the company is $N = a + b + c$, and

$$\mathbf{A}^n \begin{pmatrix} a \\ t \\ d \end{pmatrix} = \begin{pmatrix} 9/19 (a + b + c) \\ 91/209 (a + b + c) \\ 1/11 (a + b + c) \end{pmatrix} = \begin{pmatrix} (9/19)N \\ (91/209)N \\ (1/11)N \end{pmatrix}$$

So the proportions of the cars at airport, train terminal and downtown are $9/19, 91/209$ and $1/11$ respectively.

LQ4

The SDE

$$\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} 4 & -5 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

gives a direction field that describes the force field (on the phase plane). Any particle that is placed on the plane will trace out a trajectory according to the force field.

- (a) [3 marks] Write down the general solution of the SDE.
- (b) [3 marks] Sketch the phase portrait of the SDE, indicating the axes, if any, and the directions of the trajectories.
- (c) [3 marks] Suppose a particle is at the position $(2.9, 2.6)$ on the plane at time $t = 0$, write down the particular solution of the SDE that corresponds to the trajectory of this particle under the force field. (Give your answers in fractions.)
- (d) [3 marks] Is there any position (other than the origin) on the plane that we can place a particle so that it will get closer and closer towards the origin? Justify your answer.

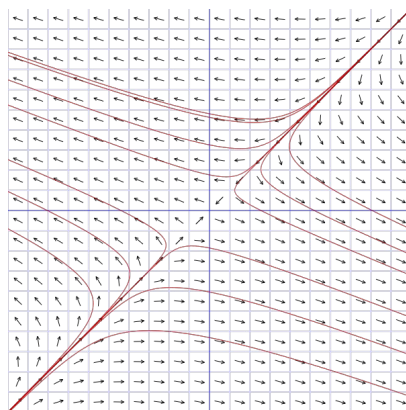
- (a) Using MATLAB, We find the eigenvalues and eigenvectors of $\begin{pmatrix} 4 & -5 \\ -2 & 1 \end{pmatrix}$:

$$r_1 = -1 \text{ with } \mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \text{ and } r_2 = 6 \text{ with } \mathbf{v}_2 = \begin{pmatrix} -5/2 \\ 1 \end{pmatrix}.$$

So the general solution of the SDE:

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{-t} + c_2 \begin{pmatrix} -5/2 \\ 1 \end{pmatrix} e^{6t}$$

- (b) The equilibrium point is a saddle, and the phase portrait has two axes determined by the two eigenvectors $\mathbf{v}_1$ and $\mathbf{v}_2$. The trajectories will start parallel to $\mathbf{v}_1$ and end parallel to $\mathbf{v}_2$.



- (c) With the initial condition, we have:

$$\begin{pmatrix} 2.9 \\ 2.6 \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} -5/2 \\ 1 \end{pmatrix}$$

Solving the system gives $c_1 = \frac{94}{35}$ and $c_2 = -\frac{3}{35}$. So the particular solution is:

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \frac{94}{35} \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{-t} - \frac{3}{35} \begin{pmatrix} -5/2 \\ 1 \end{pmatrix} e^{6t}$$

- (d) Yes. Any point along the axis determined by the eigenvector $\mathbf{v}_1$ lies on a straight line trajectory that is converging to the origin.